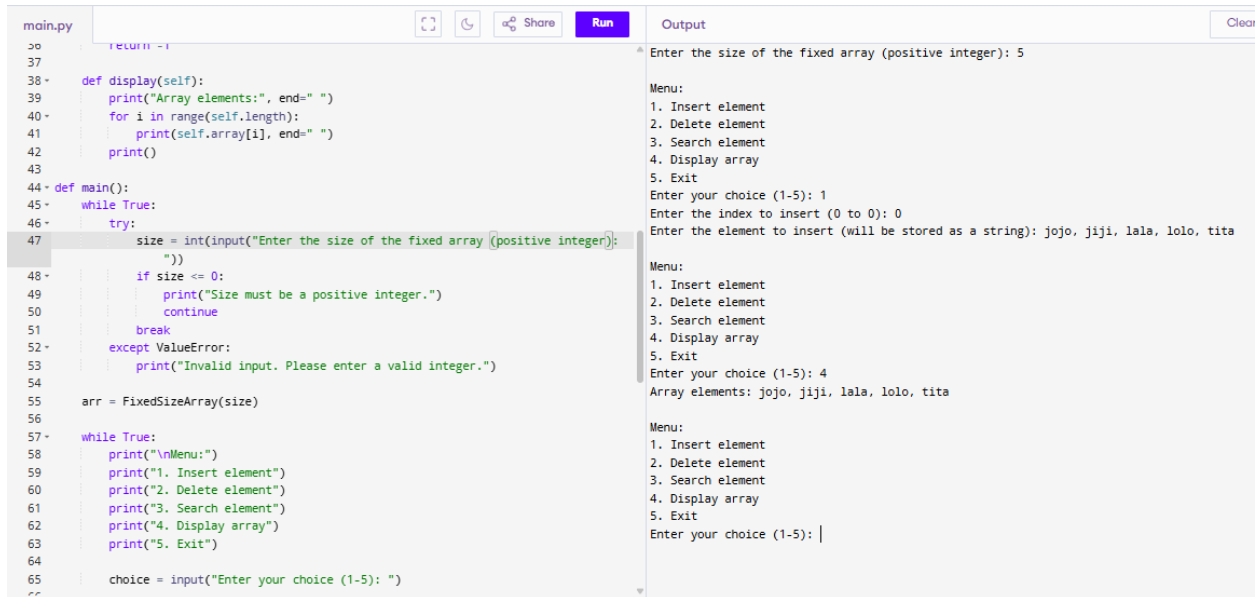


Number 2

Screenshot:



The screenshot shows a Python IDE with a file named 'main.py'. The code defines a 'FixedSizeArray' class with methods for displaying, inserting, deleting, and searching elements. The 'main' function prompts the user to enter the size of the array (5), then enters a loop where it displays a menu and processes user choices. The output window shows the execution flow: entering size 5, displaying the menu, choosing to insert, entering index 0 and element 'jojo', displaying the array, and then choosing to exit.

```
main.py
36 return -1
37
38 def display(self):
39     print("Array elements:", end=" ")
40     for i in range(self.length):
41         print(self.array[i], end=" ")
42     print()
43
44 def main():
45     while True:
46         try:
47             size = int(input("Enter the size of the fixed array (positive integer): "))
48             if size <= 0:
49                 print("Size must be a positive integer.")
50                 continue
51             break
52         except ValueError:
53             print("Invalid input. Please enter a valid integer.")
54
55     arr = FixedSizeArray(size)
56
57     while True:
58         print("\nMenu:")
59         print("1. Insert element")
60         print("2. Delete element")
61         print("3. Search element")
62         print("4. Display array")
63         print("5. Exit")
64         choice = input("Enter your choice (1-5): ")
65
66
Output
Enter the size of the fixed array (positive integer): 5
Menu:
1. Insert element
2. Delete element
3. Search element
4. Display array
5. Exit
Enter your choice (1-5): 1
Enter the index to insert (0 to 0): 0
Enter the element to insert (will be stored as a string): jojo, jiji, lala, lolo, tita
Menu:
1. Insert element
2. Delete element
3. Search element
4. Display array
5. Exit
Enter your choice (1-5): 4
Array elements: jojo, jiji, lala, lolo, tita
Menu:
1. Insert element
2. Delete element
3. Search element
4. Display array
5. Exit
Enter your choice (1-5): |
```

Programiz Link:

<https://www.programiz.com/online-compiler/000MQb2eHKcTu>

Code:

```
class FixedSizeArray:
```

```
    def __init__(self, size):
```

```
        self.size = size
```

```
        self.array = [None] * size
```

```
        self.length = 0 # to keep track of number of elements
```

```
    def insert(self, index, element):
```

```
        if self.length == self.size:
```

```
            print("Array is full. Cannot insert.")
```

```
        return

    if index < 0 or index > self.length:
        print("Invalid index.")
        return

    for i in range(self.length, index, -1):
        self.array[i] = self.array[i-1]

    self.array[index] = element
    self.length += 1

def delete(self, index):
    if index < 0 or index >= self.length:
        print("Invalid index.")
        return

    for i in range(index, self.length - 1):
        self.array[i] = self.array[i+1]

    self.array[self.length - 1] = None
    self.length -= 1

def search(self, element):
    for i in range(self.length):
        if self.array[i] == element:
            return i
```

```
return -1
```

```
def display(self):
```

```
    print("Array elements:", end=" ")
```

```
    for i in range(self.length):
```

```
        print(self.array[i], end=" ")
```

```
    print()
```

```
def main():
```

```
    while True:
```

```
        try:
```

```
            size = int(input("Enter the size of the fixed array (positive integer): "))
```

```
            if size <= 0:
```

```
                print("Size must be a positive integer.")
```

```
                continue
```

```
            break
```

```
        except ValueError:
```

```
            print("Invalid input. Please enter a valid integer.")
```

```
arr = FixedSizeArray(size)
```

```
while True:
```

```
    print("\nMenu:")
```

```
    print("1. Insert element")
```

```
    print("2. Delete element")
```

```
    print("3. Search element")
```

```
print("4. Display array")
```

```
print("5. Exit")
```

```
choice = input("Enter your choice (1-5): ")
```

```
if choice == '1':
```

```
    try:
```

```
        index = int(input(f"Enter the index to insert (0 to {arr.length}): "))
```

```
        if index < 0 or index > arr.length:
```

```
            print("Invalid index.")
```

```
            continue
```

```
        element = input("Enter the element to insert (will be stored as a string): ")
```

```
        # Optional: convert to int if you want numbers
```

```
        # element = int(element)
```

```
        arr.insert(index, element)
```

```
    except ValueError:
```

```
        print("Invalid input. Index must be an integer.")
```

```
elif choice == '2':
```

```
    try:
```

```
        index = int(input(f"Enter the index to delete (0 to {arr.length - 1}): "))
```

```
        arr.delete(index)
```

```
    except ValueError:
```

```
        print("Invalid input. Index must be an integer.")
```

```
elif choice == '3':
```

```
    element = input("Enter the element to search (stored as a string): ")
```

```
    idx = arr.search(element)
```

```
    if idx != -1:
        print(f"Element found at index {idx}.")
    else:
        print("Element not found.")
elif choice == '4':
    arr.display()
elif choice == '5':
    print("Exiting...")
    break
else:
    print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()
```